Candidate's name._____**Sign** ____

Personal number _____

UACE CHEM P 2 **SET 8**

EDUCATOR'S FORUM

Uganda Advanced Certificate of Education

SET EIGHT

CHEMISTRY

Paper 2

(Practical)

3 hours 15 minutes

INSTRUCTIONS TO CANDIDATES

- This examination paper consists of two compulsory examination items.
- Responses to these items must be written in the answer booklet provided.
- Begin each item on a fresh page.
- All working must be clearly shown. Graph papers will be provided.
- Mathematical tables and silent non-programmable scientific calculators may be used.
- You are not allowed to use reference books (i.e., textbooks, booklets on qualitative and quantitative analysis, etc.)

COMPETENCE						
ITEM 1	P	M	Dp	Da	C&R	
ITEM 2	P	O	D	C&R		

Item 1:

Inorganic salt analysis is used in mineralogy and industrial quality control to identify the cations and anions in mineral ores and industrial effluents. A mining exploration team in Eastern Uganda collected an unknown crystalline solid mineral sample, **P**, from a deposit suspected to contain iron compounds and associated transition-metal salts. At the regional laboratory the container holding P lost its identification tag.

The laboratory supervisor asked a team of Advanced Level chemistry students to carry out a systematic qualitative analysis of P. The aim is to establish which cation(s) and anion(s) it contains and to identify the compound, so that an accurate safety and composition label can be printed before industrial processing.

You are provided with about **2.0 g of solid P** in a labelled specimen bottle.

Task

1. Design and carry out a systematic qualitative analysis of P to identify the cation(s) and anion(s) present. Carry out **at least 8 distinct tests**, each with a clear observation and a logical deduction. Record your work in a table with three columns:

Test	Observation	Deduction

2. From your findings, write the **chemical formula** and the **IUPAC name** of P. Explain how it is likely to behave in water and what that means for the environment.

Item 2:

Antacids neutralise excess stomach acid, and calcium carbonate is a common active ingredient. Calcium carbonate is only sparingly soluble in water, so titrating it directly with a standard acid is slow and inaccurate. A **back titration** is used instead: an excess of standard hydrochloric acid is added to a weighed sample, and the acid that is left over is titrated with standard sodium hydroxide.

A pharmaceutical quality control laboratory received a batch of chewable antacid tablets labelled "Calci-Neutral". The quality assurance manager wants the percentage purity of calcium carbonate in the tablets, but the junior technicians need a standard procedure and calculation model.

You are provided with

Label	Substance	Given as	Concentration or amount
FA1	Hydrochloric acid, HCl (Mr = 36.5)	7.3 g dm^{-3}	**0.20 mol dm^{-3}**
FA2	Sodium hydroxide, NaOH (Mr = 40)	4.0 g dm^{-3}	**0.10 mol dm^{-3}**
Z	Crushed antacid tablet containing calcium carbonate, $CaCO_3$ (Mr = 100)	Solid	**1.25 g** in a stoppered weighing bottle
	Phenolphthalein indicator	Dropper bottle	A few drops per titration

You are also given a 250 cm^3 volumetric flask, a 25.0 cm^3 pipette, a 50.0 cm^3 pipette, a 50 cm^3 burette and conical flasks.

Task

1. Design and carry out a quantitative back-titration experiment to find the **percentage purity of calcium carbonate in solid Z**.
2. Record your titration data in a table. Show **at least six accurate titre values** (use the layout below). Show every step of your calculation, including the equations for the reactions involved.

Titration	Initial burette reading (cm^3)	Final burette reading (cm^3)	Titre of FA2 (cm^3)
Rough			
1			
2			
3			
4			
5			
6			

Mean titre = _____ cm^3 (use only titres that agree within ±0.10 cm^3)

END

EF

CHEM P 2 SET 8 CONFIDENTIAL

CONFIDENTIAL INFORMATION FOR THE HEAD OF CHEMISTRY / LABORATORY TECHNICIAN

This document contains information vital for the preparation of candidates for the Chemistry Practical examination. To ensure integrity and valid results, the contents of this document **MUST NOT** be divulged directly or indirectly to candidates.

REQUIREMENTS FOR CANDIDATES

Item 1: solid P

Item	Quantity	Notes
Solid P	About 2.0 g	Ammonium iron(II) sulfate hexahydrate, $FeSO_4 \cdot (NH_4)_2SO_4 \cdot 6H_2O$ (Mr = 392), in a clean, dry, labelled specimen bottle

Reagents for the qualitative tests are not listed in the original paper. The concentrations below are suggested standard strengths; place them on the shared bench, about 10 cm^3 of each per candidate.

Reagent	Suggested concentration
Dilute hydrochloric acid	2.0 mol dm^{-3}
Dilute nitric acid	2.0 mol dm^{-3}
Dilute sulfuric acid	1.0 mol dm^{-3}
Sodium hydroxide solution	2.0 mol dm^{-3}
Aqueous ammonia	2.0 mol dm^{-3}
Barium chloride solution	0.5 mol dm^{-3}
Lead(II) nitrate solution	0.1 mol dm^{-3}
Potassium thiocyanate solution	0.1 mol dm^{-3}
Potassium hexacyanoferrate(III) solution	0.1 mol dm^{-3}
Potassium manganate(VII) solution	0.02 mol dm^{-3}
Hydrogen peroxide solution	3% (10-volume)

Item 2:

Item	Quantity per candidate	Preparation
FA1: HCl	300 cm^3 at **0.20 mol dm^{-3}** (7.3 g dm^{-3})	Dilute **16.7 cm^3** of concentrated HCl (about 37% w/w, about 12 mol dm^{-3}) to 1.00 dm^3. Check the strength by titration before the exam.
FA2: NaOH	250 cm^3 at **0.10 mol dm^{-3}** (4.0 g dm^{-3})	Dissolve 4.00 g of NaOH in water and make up to 1.00 dm^3. Prepare fresh, because NaOH absorbs CO_2 from the air.
Solid Z: $CaCO_3$	**1.25 g** (±0.01 g)	Dry, pure $CaCO_3$ powder in a stoppered weighing bottle. Write the exact mass on the bottle.
Phenolphthalein	One dropper bottle	About 1 g in 100 cm^3 of 50% ethanol
Apparatus	One of each	50 cm^3 burette, 25.0 cm^3 pipette, 50.0 cm^3 pipette, 250 cm^3 volumetric flask, two or three 250 cm^3 conical flasks